

Glomerular Disease: Nephritic, Nephrotic, and Systemic Kidney Disease

Original Medvale study chapter. Uploaded renal disease material was used as a coverage signal; current guideline sources were used to cross-check clinical framing. No source images or tables are reproduced.

1. Why this matters clinically

Glomerular disease is high-yield because a small cluster of findings can narrow a large differential quickly. Urine microscopy, protein quantity, serum albumin, creatinine trajectory, blood pressure, edema, complement levels, and serologies convert vague kidney dysfunction into a recognizable syndrome. The stakes are high: rapidly progressive glomerulonephritis can destroy kidney function over days to weeks, while nephrotic syndrome increases risks of thrombosis, infection, dyslipidemia, and progressive chronic kidney disease.

The first move is to classify the urine pattern. Nephritic disease is an inflammatory capillary-wall problem, so hematuria, dysmorphic red cells, RBC casts, hypertension, and falling GFR dominate. Nephrotic disease is a permeability problem, often involving podocytes or the basement membrane, so heavy albuminuria, edema, hypoalbuminemia, hyperlipidemia, and lipiduria dominate. Many diseases are mixed, especially lupus nephritis and membranoproliferative patterns.

Diagnostic shortcut: Pattern -> Tempo -> Complement -> Systemic clues -> Biopsy decision. Pattern identifies the syndrome, tempo identifies urgency, complement identifies immune pathway, systemic clues identify disease family, and biopsy identifies histology when treatment depends on it.

2. Core illness scripts

Syndrome	Urine and serum pattern	Clinical image	First pivot
Nephritic	Hematuria, dysmorphic RBCs/RBC casts, rising creatinine, hypertension, protein often <3.5 g/day	Inflamed glomerular capillary wall leaks cells and reduces filtration	Check C3/C4, ANA/dsDNA, ANCA, anti-GBM, hepatitis/cryoglobulins; biopsy if severe or progressive
Nephrotic	Protein >3.5 g/day, low albumin, edema, hyperlipidemia, lipiduria	Podocyte or basement membrane injury allows albumin loss	Quantify protein; evaluate diabetes, PLA2R, infections, malignancy, monoclonal disease; biopsy if unexplained
RPGN	Nephritic urine plus rapid GFR loss over days-weeks	Crescents compress capillary tufts	Urgent nephrology; serologies and biopsy; disease-specific immunosuppression
Mixed	Hematuria/casts plus heavy proteinuria	Inflammation and permeability coexist	Think lupus, MPGN/C3, infection-related GN, IgA variants

3. Pathophysiology of the glomerular filter

The filtration barrier has fenestrated endothelium, glomerular basement membrane, and podocyte slit diaphragms. Nephrotic disease primarily reflects increased permeability to albumin. Nephritic disease primarily reflects immune or inflammatory injury to capillary walls, allowing red cells to enter urine while decreasing filtration surface area. Biopsy terms can describe mechanisms or patterns: membranous nephropathy implies immune deposits along the capillary wall; focal segmental glomerulosclerosis can be a primary podocytopathy or a scar pattern; membranoproliferative glomerulonephritis is a pattern that may arise from immune complexes, monoclonal proteins, infection, lupus, or alternative-complement dysregulation.

4. Diagnostic approach

Confirm the signal before naming the disease. Repeat a contaminated sample; obtain microscopy; quantify protein with urine protein-to-creatinine ratio, albumin-to-creatinine ratio, or 24-hour urine if needed. A dipstick is albumin-weighted and may

under-detect light chains, so paraprotein concerns require serum and urine protein electrophoresis and serum free light chains.

Initial data usually include creatinine/eGFR trend, serum albumin, lipid profile, urinalysis with microscopy, urine protein quantification, blood pressure, edema exam, C3/C4, ANA and anti-dsDNA when lupus is plausible, ANCA, anti-GBM, hepatitis B/C and HIV testing, and monoclonal protein testing when appropriate. Ultrasound evaluates kidney size and obstruction but rarely classifies glomerular disease. Biopsy is most useful when the result changes treatment.

Biopsy is commonly considered for unexplained adult nephrotic syndrome, active sediment with falling GFR, suspected rapidly progressive GN, suspected lupus nephritis, and atypical diabetic kidney disease. Atypical diabetic features include rapid creatinine rise, RBC casts, low complement, abrupt nephrotic syndrome, short diabetes duration, or absent retinopathy.

5. Nephritic syndromes

Nephritic disease is inflammation at the capillary wall. Patients may present with cola-colored urine, RBC casts, hypertension, oliguria, edema, and AKI. Tempo is critical. Hematuria concurrent with a respiratory infection suggests IgA nephropathy. Hematuria weeks after skin or throat infection with low C3 suggests post-infectious GN. Pulmonary hemorrhage with rapidly rising creatinine suggests ANCA vasculitis or anti-GBM disease.

Disease	Classic clues	Complement/serology	Anchor
IgA nephropathy	Hematuria within 1-2 days of URI; recurrent microscopic hematuria	Usually normal C3/C4	Mesangial IgA; supportive care and risk-based nephrology therapy
Post-infectious GN	Hematuria, edema, HTN after skin/throat infection	Low C3; strep serologies may help	Treat infection/supportive care; biopsy if atypical or progressive
Lupus nephritis	SLE symptoms, proteinuria, casts, cytopenias	Low C3/C4; ANA, anti-dsDNA	Biopsy class guides therapy
ANCA GN	RPGN, ENT/lung/skin/nerve findings	MPO/p-ANCA or PR3/c-ANCA; often normal complement	Pauci-immune necrotizing GN; urgent induction therapy
Anti-GBM disease	Pulmonary hemorrhage + GN; rapid course	Anti-GBM antibodies; linear IgG	Time-sensitive plasma exchange/immunosuppression in selected cases
Cryoglobulinemic GN	Purpura, arthralgia, neuropathy, HCV or lymphoproliferative disease	Very low C4; RF; cryoglobulins	Immune-complex MPGN pattern; treat trigger and severe disease

6. Complement pattern shortcut

Pattern	Likely mechanism	Prioritize	Next tests
Normal complement	IgA, pauci-immune, anti-GBM, hereditary GBM disorders	IgA nephropathy, ANCA GN, anti-GBM	ANCA, anti-GBM, microscopy, protein quantification, biopsy if active/progressive
Low C3 only	Alternative pathway or post-infectious pattern	Post-infectious GN, C3 glomerulopathy	Infection evaluation, repeat complement, biopsy if persistent/progressive
Low C3 + low C4	Classical immune-complex activation	Lupus, cryoglobulinemia, immune-complex MPGN, endocarditis	ANA/dsDNA, cryoglobulins, RF, hepatitis tests, blood cultures if infection clues
Very low C4	Cryoglobulins or classical pathway dominance	HCV cryoglobulinemia, lymphoproliferative disease	Cryoglobulin handling, HCV, SPEP/free light chains

7. Nephrotic syndromes

Nephrotic syndrome is heavy protein loss with systemic consequences. The adult threshold is classically >3.5 g/day of proteinuria. Albumin loss lowers oncotic pressure, but kidney sodium retention also contributes to edema. Complications include venous thrombosis, serious infection, severe dyslipidemia, vitamin D-binding protein loss, altered drug binding, and progression to CKD.

Disease	Typical clue	Diagnostic association	Management anchor
Minimal change disease	Child with edema after URI; adult cases with NSAIDs or Hodgkin lymphoma	Light microscopy often near-normal; foot process effacement	Steroid-responsive pattern; evaluate secondary causes in adults
FSGS	Obesity, HIV, heroin, reduced nephron mass, reflux, APOL1 risk variants	Segmental sclerosis in some glomeruli	Primary vs secondary distinction drives immunosuppression vs supportive care
Membranous nephropathy	Adult nephrotic syndrome; malignancy, HBV, autoimmune disease, drugs	Anti-PLA2R supports primary disease	Risk stratify; screen secondary causes; immunosuppression for higher-risk disease
Diabetic kidney disease	Long diabetes duration, retinopathy, gradual albuminuria	Nodular sclerosis can occur	Glycemic/BP control, RAAS blockade if tolerated, SGLT2 inhibitor if eligible
Amyloid / monoclonal disease	Nephrotic proteinuria with older age, HFpEF, neuropathy, plasma cell clues	Congo red amyloid or monoclonal deposits	Identify type/clone; hematology-nephrology coordination

8. Systemic kidney disease patterns

Systemic features can make the glomerular diagnosis easier. Lupus adds rash, arthritis, cytopenias, low complements, and anti-dsDNA activity. ANCA vasculitis adds ENT disease, pulmonary nodules or hemorrhage, neuropathy, purpura, and constitutional symptoms. Anti-GBM disease adds pulmonary hemorrhage and a very rapid renal course. Cryoglobulinemia adds palpable purpura, neuropathy, hepatitis C, and very low C4. Diabetic kidney disease adds retinopathy, neuropathy, longstanding hyperglycemia, and gradual albuminuria.

9. Management principles

Supportive care is the floor for almost every glomerular disease: blood pressure control, proteinuria reduction, sodium restriction for edema, avoidance of nephrotoxins, cardiovascular risk reduction, and vaccination planning when immunosuppression is likely. RAAS blockade is common in proteinuric disease when tolerated, but creatinine and potassium must be monitored. SGLT2 inhibitors are part of modern proteinuric CKD care for many eligible patients, including some without diabetes, but acute illness, ketoacidosis risk, pregnancy, and eGFR thresholds require individualized decisions.

Disease-specific therapy depends on the mechanism. Minimal change disease is often steroid responsive. Membranous nephropathy ranges from monitoring to immunosuppression depending on risk. Primary FSGS may require immunosuppression, but secondary FSGS is treated by reducing glomerular stress. Lupus nephritis therapy depends on biopsy class and activity. ANCA vasculitis and anti-GBM disease are inflammatory emergencies. Infection-related GN requires source control and careful avoidance of reflexive immunosuppression unless selected severe circumstances exist.

Problem	Initial response	Why
Nephrotic edema	Sodium restriction plus loop diuretic strategy; monitor weight, creatinine, K/Mg	Overdiuresis can worsen AKI; persistent overload increases morbidity
Proteinuria	RAAS blockade when tolerated; BP control; SGLT2 inhibitor if eligible	Proteinuria predicts and promotes progression
Thrombosis risk	Assess albumin, membranous disease, prior VTE, bleeding risk	Nephrotic syndrome shifts toward hypercoagulability
RPGN	Urgent nephrology, serologies, biopsy planning, disease-specific therapy	Delay converts reversible inflammation to fibrosis

10. Differentials and mimics

Red urine is not always glomerular hematuria. Myoglobinuria and hemoglobinuria can produce positive dipstick blood with few or no RBCs. Stones, tumors, infection, catheter trauma, exercise, and menstruation can produce hematuria without dysmorphic RBCs or casts. Similarly, protein-positive dipstick is not always nephrotic disease, while light chain disease may have severe kidney injury with only modest dipstick positivity.

Hypertensive emergency can mimic glomerulonephritis with hematuria, proteinuria, and AKI; glomerulonephritis can also cause severe hypertension. Diabetes is common, but rapid decline, active sediment, sudden nephrotic syndrome, low complement, or absent retinopathy should trigger evaluation for non-diabetic kidney disease.

11. High-yield exam clues

Clue	Think
Gross hematuria immediately after URI	IgA nephropathy
Hematuria weeks after impetigo/pharyngitis + low C3	Post-infectious GN
RBC casts + pulmonary hemorrhage	ANCA vasculitis or anti-GBM disease
Low C4 + purpura + neuropathy + hepatitis C	Cryoglobulinemic GN
Adult nephrotic syndrome + anti-PLA2R	Primary membranous nephropathy
Child with selective proteinuria and steroid response	Minimal change disease
Obesity/HIV/reduced nephron mass + proteinuria	Secondary FSGS
Dipstick blood positive but no RBCs	Myoglobinuria or hemoglobinuria

12. Common pitfalls

- Naming disease from protein quantity alone. Heavy proteinuria defines severity, not etiology.

- Calling hematuria glomerular without microscopy. RBC casts and dysmorphic RBCs matter.
- Assuming all diabetic CKD is diabetic nephropathy despite atypical findings.
- Ignoring complement recovery after infection-related GN. Persistently low C3 is a warning sign.
- Waiting too long in RPGN. Rapid creatinine rise with active sediment is time-sensitive.
- Forgetting nephrotic complications: thrombosis, infection, severe dyslipidemia, and drug-binding changes.

13. Practice questions

1. A 23-year-old develops tea-colored urine one day after a URI. BP is mildly elevated, creatinine is stable, and complements are normal. Most likely diagnosis?

Answer: IgA nephropathy: synpharyngitic hematuria and normal complement are key.

2. A child develops edema and cola-colored urine three weeks after impetigo. C3 is low. What is the first management principle?

Answer: Supportive care and treatment of active infection if present; biopsy if atypical or progressive.

3. An adult has anasarca, albumin 2.0 g/dL, proteinuria 8 g/day, and anti-PLA2R antibodies. Diagnosis suggested?

Answer: Primary membranous nephropathy, with secondary cause evaluation and risk stratification.

4. Hemoptysis, rapidly rising creatinine, RBC casts, and anti-GBM antibodies. Why urgent?

Answer: Anti-GBM disease can cause pulmonary hemorrhage and irreversible RPGN.

5. Hepatitis C, palpable purpura, neuropathy, renal dysfunction, and very low C4. Likely renal pattern?

Answer: Cryoglobulinemic immune-complex GN, often with an MPGN pattern.

6. Diabetic patient develops abrupt nephrotic syndrome, active sediment, and no retinopathy. Trap?

Answer: Do not assume diabetic kidney disease; evaluate for non-diabetic glomerular disease.

14. Graph-RAG extraction map

Node	Classes	Key edges
Nephritic syndrome	Renal syndrome; inflammatory glomerular disease	causes hematuria; associated_with RBC casts; may_progress_to RPGN
Nephrotic syndrome	Renal syndrome; permeability disease	defined_by proteinuria >3.5 g/day; causes edema; increases_risk thrombosis/infection
IgA nephropathy	Immune-complex GN	triggered_by mucosal infection; has_complement normal
Post-infectious GN	Low-complement GN	follows infection; lowers C3; treated_by supportive care
Lupus nephritis	Systemic autoimmune kidney disease	associated_with SLE; lowers C3/C4; biopsy_guides therapy
ANCA-associated GN	Pauci-immune vasculitis; RPGN	associated_with pulmonary-renal syndrome; diagnosed_by ANCA/biopsy
Anti-GBM disease	Pulmonary-renal syndrome; RPGN	caused_by anti-GBM antibodies; associated_with linear IgG
Membranous nephropathy	Nephrotic disease	associated_with PLA2R; increases_risk renal vein thrombosis
FSGS	Podocyte injury/scar pattern	can_be primary or secondary; associated_with obesity/HIV/reduced nephron mass
Complement pattern	Diagnostic axis	stratifies immune pathway and differential diagnosis

15. References used for clinical cross-checking

- KDIGO 2021 Clinical Practice Guideline for the Management of Glomerular Diseases.
- KDIGO 2024 Clinical Practice Guideline for the Management of Lupus Nephritis.
- KDIGO 2024 Clinical Practice Guideline for ANCA-Associated Vasculitis.
- KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of CKD.
- Uploaded renal disease source packet used as a coverage signal; source tables and figures were not reproduced.

16. Disease deep dives for illness-script retention

Minimal change disease is a podocyte foot-process effacement syndrome. In children, the classic presentation is abrupt edema, selective albuminuria, and preserved complement after a viral illness or allergic trigger. Adults require more caution because MCD can be associated with NSAIDs, Hodgkin lymphoma, and other secondary causes. The serum creatinine can

be normal or elevated from intravascular depletion, severe edema physiology, or acute tubular injury. The conceptual anchor is that the glomerulus looks deceptively normal by light microscopy while the filtration slit diaphragm is functionally leaky.

FSGS is best understood as a pattern plus a question: primary or secondary? Primary FSGS behaves like a circulating-factor podocytopathy and may present with abrupt nephrotic syndrome. Secondary FSGS is glomerular scarring from adaptive stress, such as obesity, reduced nephron mass, reflux nephropathy, longstanding hypertension, HIV, or drug exposure. This distinction matters because primary FSGS may be treated with immunosuppression, while secondary FSGS is treated by reducing glomerular pressure and removing the stressor.

Membranous nephropathy is a classic adult nephrotic syndrome. The clinical script is edema, heavy proteinuria, and sometimes renal vein thrombosis. Anti-PLA2R antibodies strongly support primary membranous nephropathy, but the clinician should still consider age-appropriate malignancy screening and secondary causes such as hepatitis B, autoimmune disease, and medications. Risk is estimated from proteinuria severity, eGFR, serum albumin, antibody trend, and complications rather than protein level alone.

IgA nephropathy is a mucosal immune-complex disease. The memorable timing is synpharyngitic hematuria: gross hematuria appears during or within a day or two of an upper respiratory infection. Complement is usually normal. The course ranges from benign recurrent hematuria to progressive CKD, especially when proteinuria, hypertension, or reduced GFR are present.

Lupus nephritis is a kidney manifestation of systemic immune-complex disease. Serologies can suggest activity, but biopsy class determines the treatment pathway because class II mesangial disease, class III/IV proliferative disease, class V membranous disease, and mixed patterns carry different risks. Serial proteinuria, creatinine, urine sediment, complement, and anti-dsDNA trends are useful, but none substitute for histology when therapy decisions are high-stakes.

17. Biopsy pattern vocabulary

Biopsy word	Meaning	Clinical translation
Focal	Some glomeruli involved	Disease may be missed if sample is small; focal does not mean mild
Diffuse	Most glomeruli involved	Usually larger disease burden
Segmental	Part of an individual glomerulus involved	Classic in FSGS and some necrotizing lesions
Global	Entire glomerulus involved	Often more advanced or chronic damage
Crescent	Proliferating cells and fibrin in Bowman space	Marker of severe capillary-wall injury and RPGN risk
Spike and dome	Basement membrane reaction around subepithelial deposits	Classic teaching phrase for membranous nephropathy
Linear IgG	Smooth antibody staining along GBM	Anti-GBM disease
Granular deposits	Lumpy immune-complex deposition	Lupus, post-infectious, membranous, IgA, MPGN patterns
Pauci-immune	Little or no immune staining	ANCA-associated necrotizing GN

18. Original diagnostic algorithm in prose

Start with the urinalysis. If dipstick blood is positive, compare microscopy. Many RBCs with dysmorphic forms or RBC casts suggests glomerular hematuria. Positive blood with few RBCs suggests pigment such as myoglobin or hemoglobin. If protein is present, quantify it. Nephrotic-range protein moves the differential toward MCD, FSGS, membranous nephropathy, diabetic kidney disease, amyloid, and mixed lupus or MPGN patterns. Subnephrotic protein with RBC casts moves the differential toward nephritic disease.

Next determine acuity. A stable creatinine allows outpatient nephrology evaluation in many cases, but a rapidly rising creatinine with active urine sediment is a renal emergency. Then use complement as the branching variable. Low C3 alone suggests post-infectious GN or C3 glomerulopathy. Low C3 and C4 suggests classical immune-complex disease such as lupus, cryoglobulinemia, endocarditis-associated GN, or immune-complex MPGN. Normal complement keeps IgA nephropathy, ANCA vasculitis, anti-GBM disease, and non-inflammatory hereditary basement membrane disease on the table.

Finally ask whether the disease is kidney-limited or systemic. Pulmonary hemorrhage, sinus disease, purpura, neuropathy, rash, arthritis, fever, cytopenias, hepatitis risk, malignancy symptoms, and monoclonal protein clues should redirect testing. Biopsy is the final classifier when treatment depends on histology or when clinical data conflict.

19. Treatment safety anchors

Safety issue	Why it matters	Practical anchor
RAAS blockade	Reduces intraglomerular pressure and proteinuria but can raise creatinine/K	Check creatinine and potassium after starting or increasing dose
Diuretics in nephrotic edema	Edema may coexist with low effective arterial volume	Track daily weight, urine output, orthostasis, creatinine, K/Mg

Immunosuppression	Infection, malignancy, bone, glucose, and fertility risks vary by agent	Vaccination and prophylaxis planning matter before therapy
Anticoagulation	Nephrotic syndrome increases thrombosis risk but bleeding risk varies	Individualize based on albumin, membranous disease, VTE history, bleeding risk
RPGN delay	Crescents can become irreversible fibrosis	Early nephrology/biopsy planning is part of treatment, not an optional add-on
Infection-related GN	Immunosuppression can worsen uncontrolled infection	Prioritize source control; biopsy if atypical or progressive

20. Additional integrated cases

1. Case 1: A 67-year-old with weight loss and nephrotic syndrome has negative anti-PLA2R.

Answer: Think membranous nephropathy with secondary-cause evaluation, including malignancy screening guided by age and symptoms.

2. Case 2: A patient with obesity and long-standing hypertension has proteinuria of 2 g/day and bland sediment.

Answer: Think adaptive/secondary FSGS or hypertensive nephrosclerosis rather than primary nephrotic podocytopathy.

3. Case 3: A patient has sinusitis, pulmonary nodules, purpura, RBC casts, and positive PR3-ANCA.

Answer: Think granulomatosis with polyangiitis causing pauci-immune crescentic GN.

4. Case 4: A patient has nephrotic syndrome, orthostatic symptoms after aggressive diuresis, and rising creatinine.

Answer: Think overdiuresis or reduced effective arterial volume superimposed on nephrotic physiology; reassess volume and medications.

5. Case 5: A patient has low C3 for months after presumed post-infectious GN.

Answer: Do not keep assuming routine post-infectious GN; evaluate for C3 glomerulopathy or another persistent complement-mediated process.

6. Case 6: A patient has a weakly positive dipstick protein but severe AKI, anemia, hypercalcemia, and back pain.

Answer: Think light chain-related kidney disease; dipstick can under-detect non-albumin proteins.

21. Graph-RAG relation labels worth preserving

Relationship label	Example edge	Activation cue
presents_with	IgA nephropathy -> gross hematuria	Hematuria close to URI
lowers	Lupus nephritis -> C3/C4	Low complement plus SLE features
associated_with	Membranous nephropathy -> renal vein thrombosis	Nephrotic syndrome plus flank pain/hematuria
mimics	Hypertensive emergency -> nephritic urine	Severe BP plus AKI/hematuria
requires_urgent_evaluation	RPGN -> nephrology/biopsy	Rapid creatinine rise plus active sediment
treated_by_supportive_care	Proteinuric glomerular disease -> BP/proteinuria control	Stable disease or as baseline therapy
contraindication_caution	Infection-related GN -> immunosuppression caution	Evidence of active infection